Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/21/2022
PI Name:	Gilbert, Jack Ph.D.		
Project Title:	Quantifying Selection for Pathogenicity and Antibiotic Resistance in Bacteria and Fungi on the ISS – a Microbial Tracking Study		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gilbertjack@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	630-915-2383
Organization Name:	University of California, San Diego		
PI Address 1:	9500 Gilman Dr		
PI Address 2:	MC 0763		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-5004	**Congressional District:**	52
Comments:			
Project Type:	Flight,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	10/01/2019	**End Date:**	09/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per PI and NSSC information (Ed., 9/21/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Urbaniak, Camilla Ph.D. (NASA Jet Propulsion Laboratory) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC19K1604		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbes inhabit virtually every realm of our planet. Over their 4-billion-year evolutionary history, they have shown remarkable adaptation to the environment. However, these same microbes are now adapting to our built environments, such as our homes, hospitals, offices, and closed habitats like the International Space Station (ISS). Built environment-associated bacteria and fungi originate mostly from human occupants and, as such, are often adapted to living on or in our bodies. Therefore, when we shed microbes into our buildings, they land in a relatively dry, nutrient poor environment. This environmental stress can trigger horizontal gene transfer events that facilitate the acquisition of new genes, which could ultimately result in an increased risk to human health. We hypothesize that if the built environment promotes microbial stress-survival, we will see an increase in antibiotic resistance and virulence. To test our hypothesis, we have focused on the Microbiomes of the Built Environment of Spacecraft, using existing data (GeneLab) and isolated microbial strains held at Jet Propulsion Laboratory (JPL)/NASA. We are sequencing these microbial genomes and quantifying the types and abundance of antibiotic resistance and virulence genes. Additionally, we are performing metagenomic analysis of whole communities collected from eight distinct ISS surface environments during the MT-1 and MT-2 missions, as well collecting new samples as part of this MT-3 project. This study will help us understand the impact of spaceflight stress on structural and functional changes in ISS-associated microbes over time and predict which microbial taxa are likely to pose a threat to crew health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding how bacteria and fungi adapt to the built environment and assessing associated health risks for human occupants should be of global research priority. Exploring microbial adaption on the ISS provides a unique opportunity, due to its well-controlled environmental variation and isolation from other sources of environmental microbes -- something which we cannot control for in terrestrial built environments. Therefore, while our study provides insights into potential risks for astronaut health, it also allows us to identify the environmental drivers that might alter microbial infection risk in built environments found on Earth.
Task Progress:	During the third year of our award, we successfully completed our sample collection for the second and third MT-3 flights on the SpaceX CRS-23 (SpX-23) and SpaceX CRS-24 (SpX-24) missions (Task 8). All protocol and storage requirements were met, resulting in an “Excellent” rating. Co-Investigators at NASA Jet Propulsion Laboratory (JPL) have completed all sample processing, including propidium monoazide (PMA) treatment, isolate culturing, retainment of sample aliquots for long-term storage in the NASA Life Sciences Data Archive (LSDA) repository, estimation of microbial burden with qPCR, and extraction of both isolate and metagenomic DNA for all MT-3 flights (Tasks 8 and 9). All resulting samples/data will be used to complete MT-3-associated in vitro and in vivo assays and bioinformatic analyses. For individual isolates, we have completed annotation of antibiotic resistance and virulence genes for all MT-1 and MT-2 samples (Task 3), and DNA from MT-3 isolates has been submitted for whole genome sequencing (Task 10). For metagenomic samples, we have completed assembly and annotation of MT-1 and MT-2 sequences (Task 4), and we have begun submission of MT-3 samples to the University of California San Diego (UCSD) Microbiome Core Facility for shotgun sequencing (Task 10). Further, the Principal Investigator (PI) team at UCSD has begun antibiotic resistance gene verification assays for MT-1 and MT-2 isolates. To date, we have challenged 37 International Space Station (ISS) strains with six antibiotics, including aztreonam, gemifloxacin, penicillin, levofloxacin, azithromycin, and oxacillin, with validation of results through experimental replication (Task 6). We have also begun virulence assays using a C. elegans nematode killing model, with several strains successfully tested to date (Task 7). Ed. Note: Tasks include - Task 1. Sequence whole genomes of ISS microorganisms Task 2. Generate bioinformatic antibiotic resistance profiles of ISS microorganisms Task 3. Elucidate genes associated with host-invasion potential of ISS microorganisms Task 4. Assemble genomes from metagenomic data Task 5. Predict ancestral genotypes of key bacterial species Task 6. Perform in vitro antibiotic resistance assays Task 7. Perform in vivo virulence assays using C. elegans Task 8. ISS environmental sampling Task 9. Estimate and identify culture-based microbial burden and diversity Task 10. Perform metagenomic sequencing and qPCR of PMA-treated DNA to examine the viable phylogenetic and functional diversity
Bibliography Type:	Description: (Last Updated: 11/18/2025)
Articles in Peer-reviewed Journals	Simpson AC, Suzuki T, Miller DR, Venkateswaran K. "Microbial burden estimation of food items, built environments, and the International Space Station using film media." Microorganisms. 2022 Aug 25;10(9):1714. https://doi.org/10.3390/microorganisms10091714 ; PMID: 36144316; PMCID: PMC9503880 , Aug-2022